Amirhossein Dezhboro

Ph.D. Student in Systems Engineering

Stevens Institute of Technology
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Research Interests

Machine Learning Applications in Social Sciences, Natural Language Processing, Social Network Analysis, Urban Sustainability and Resilience, Community Dynamics, Computational Social Systems, Human Behavior Analysis, Climate Impact on Urban Communities

Education

2024 – Present **Ph.D. in Systems Engineering**, *Stevens Institute of Technology*, *GPA: 4.0/4.0*.
Research Advisor: Dr. Jose Emmanuel Ramirez-Marquez
Relevant Coursework: Applied Machine Learning, Data Science and Knowledge Discovery, Decision Making Via Data Analysis Techniques

2018 – 2021 **M.Sc. in Systems Engineering**, *IUST*, *GPA: 3.77/4.0*.
Relevant Coursework: Artificial Intelligence and Expert Systems, Data Mining
Thesis: Data Mining to Discover the Community Response and Behavior at Disasters in Social Media with a Multi-dimensional Data Fusion Approach
Advisor: Dr. Mahdi Ghazanfari

Publications and Talks

Journal Articles

2025 **A. Dezhboro**, P. Babvey, C. Lipizzi, and J. E. Ramirez-Marquez. Analyzing Social Landscapes: Visualizing the Key Elements of Social Media Dynamics. *IEEE Transactions on Computational Social Systems*, doi:10.1109/TCSS.2025.3591139

2025 F. Maleki, and Saeed Yaghoubi, **A. Dezhboro**. Organic production and distribution in a two-echelon food supply chain with stochastic demand using a game theory approach. *International Journal of Systems Science: Operations & Logistics, 12(1).*, doi:10.1080/23302674.2025.2539912

2024 **A. Dezhboro**, J. E. Ramirez-Marquez, and A. Krstikj. Community Shaping in the Digital Age: A Temporal Fusion Framework for Analyzing Discourse Fragmentation in Online Social Networks. arXiv:2409.11665.

2023 H. Kalantari, A. Badiee, **A. Dezhboro**, H. Mohammadi and E. B. Tirkolaee. A Fuzzy Profit Maximization Model Using Communities Viable Leaders for Information Diffusion in Dynamic Drivers Collaboration Networks. *IEEE Transactions on Fuzzy Systems*, vol. 31, no. 2, pp. 370-379.

Conference Papers

2025 **A. Dezhboro**, R. Contreras-Galvan, J. Ramirez-Marquez. Perception-Enhanced Vulnerability Analysis: A Human-Centric Vulnerability Metric Framework Using Multimodal Social Intelligence for Disaster Response. *INFORMS*, Accepted

2020 M. Ghazanfari, H. Mohammadi, **A. Dezhboro**, and M. Khanzadi. Urban data analysis and management platform based on big data and Internet of Things for the city of Tehran. *Smart Tehran Conference*, vol. 1, issue SMARTTEH01_023, 8 pages.

Working Papers

2025 **A. Dezhboro**, J. Ramirez-Marquez. Unequal Disruptions: Measuring the Fairness of Emergency Impacts on Daily Mobility Across Socioeconomic Divides. *Working Paper*

Professional Experience

Jan 2025 – May 2025 **Teaching Assistant**, *Stevens Institute of Technology*, Hoboken, NJ.

Elements of Operations Research (EM 605-A)

- Guide students in applying linear programming and mathematical modeling techniques
- Conduct tutorial sessions on optimization methods, linear programming, and decision analysis

Project Management of Complex Systems (EM 612-A)

- Guide students in developing project plans, defining scope, and performing risk analysis.
- Conduct tutorial sessions on project scheduling, resource allocation, and budget management.

Oct 2021 – Aug 2023 **Research and Development Engineer**, *SAB*, Tehran, Iran.

- Developed high-performance reinforcement learning models for profit maximization, risk minimization, pairs trading, and strategy switching
- Applied state-of-the-art theories in portfolio management, utilizing stochastic optimization, time-series analysis, and reinforcement learning techniques
- Developed and backtested statistical models to evaluate the performance of trading strategies against diverse criteria.
- Conducted system maintenance and debugging to ensure the accuracy and reliability of analytical tools and data.

May 2017 – Sep 2018 **Game Developer**, *Butterfi Studio*, Tehran, Iran.

- Collaborated with a designer to develop indie games using C# and C++ (Unity Engine)
- Contributed to game mechanics, programming, and system design

Technical Skills

Programming Languages	Python (PyTorch, Django, Flask), C++, C#, R, JavaScript (D3.js, Vue.js), SQL (MySQL, PostgreSQL)
Machine Learning	Large Language Models, Deep Reinforcement Learning, Time-series Analysis, Signal Processing, Statistical Modeling, AutoML
Data Science	Data Visualization, Linear Programming, Optimization Models, Social Network Analysis, Natural Language Processing, Causal Inference
Tools & Frameworks	Stochastic Optimization, Portfolio Management Systems

Academic Achievement

Ph.D.	Perfect GPA (4.0/4.0) at Stevens Institute of Technology
Publications	Publishing in high impact journals like IEEE transactions on fuzzy systems, and IEEE Transactions on Computational Social Systems
Citations	Research work has received 19+ citations across multiple domains

Scientific Sociaties

2025 – Present	**IEEE Computer Society**.
2025 – Present	**IEEE Young Professionals**.
2025 – Present	**INFORMS Data Mining Society**.
2025 – Present	**INFORMS Analytics Society**.